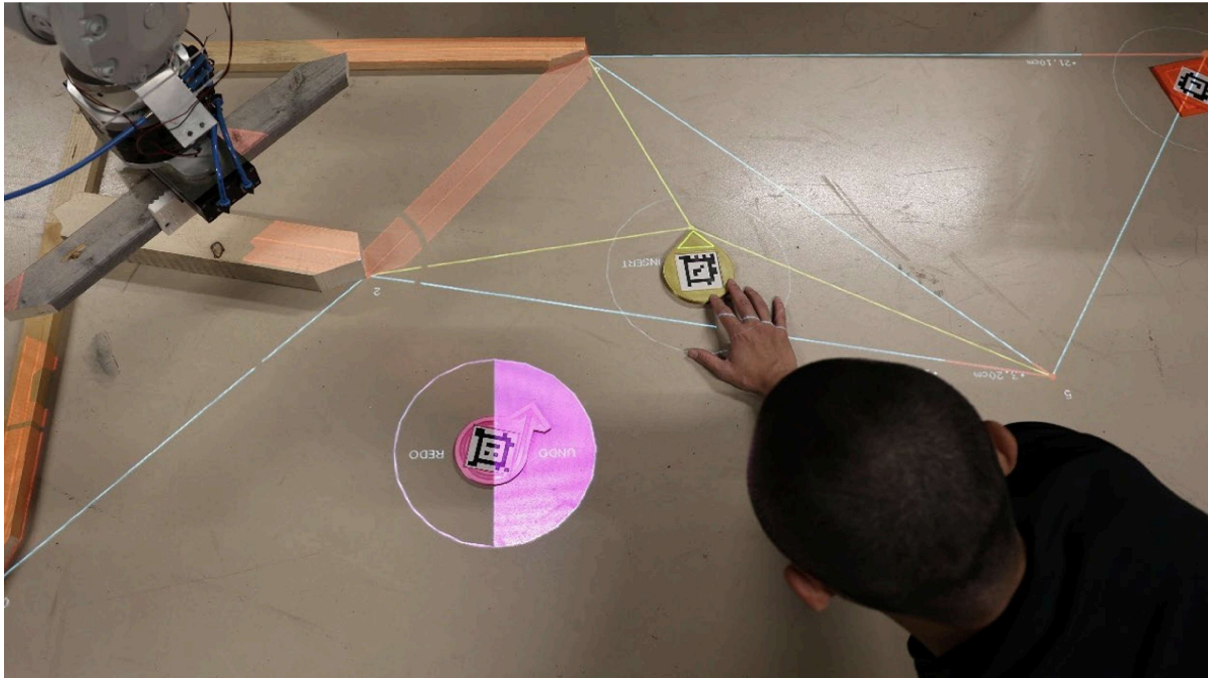


HARDWARE III

Human-in-the-Loop Interactive Systems

25.26 MRAC01 AND MAAI01



Credits: EPFL+ETH summer workshop

Syllabus

This course explores the design of human-in-the-loop interactive systems for fabrication, where human input, computational logic, and real-time feedback are integrated into a unified workflow.

The course focuses on the relationship between user input, system processing, and responsive output. Students will explore different forms of interaction—including mouse, gesture, vision-based systems, and physical interfaces—and learn how these inputs can be interpreted and translated into meaningful system behaviors.

Rather than directly generating form, users interact with systems that modulate rules, behaviors, and fabrication processes. These interactions are structured through computational frameworks such as mapping strategies, finite state machines (FSM), and behavioral logic.

Special attention is given to user experience, human-centered design, and interaction design, encouraging students to create systems that are intuitive, engaging, and spatial. The course emphasizes physical and embodied interaction, avoiding conventional screen-based paradigms and ensuring that systems do not behave like touch interfaces.

Students will develop a complete interactive system within a controlled setup, integrating input methods, system logic, and feedback mechanisms. The course culminates in the design of a rule-based system using projection mapping, where interaction guides the process through real-time visual feedback.

Learning Objectives

By the end of the course, students will be able to:

- Understand and apply the relationship between user input, system logic, and responsive output
- Design human-in-the-loop workflows for fabrication processes
- Translate different types of human interaction into computational behaviors
- Develop rule-based systems using mapping strategies and finite state machines (FSM)
- Structure interaction using state-based and behavioral logic
- Design and implement real-time feedback systems using projection and visualization
- Apply principles of user experience (UX) and human-centered design in physical interaction contexts
- Prototype interactive systems that guide processes through rules and feedback

Hardware / Software requirements

Hardware:

- Laptop (macOS or Windows)
- Video projector (HDMI connection, basic HD resolution)
- Flat white surface (table, board, or wall for projection & assembly)

Optional (for extended interaction and sensing experiments):

- USB webcam (for basic tracking and vision-based input)
- Microcontroller (e.g., Arduino or ESP32) with simple sensors (button, distance, etc.)
- Laser pointer / laser module (for alternative input or interaction techniques)

Software:

- Rhino + Grasshopper
- Python (inside Grasshopper, optional for advanced logic)
- Anemone (optional, for loop / FSM logic)
- Human UI (optional, for user interaction interface)
- Firefly (optional, for input devices / camera integration)

Optional for projection mapping / interface design (choose one):

- TouchDesigner
- MadMapper
- HeavyM

Faculty



Hamid Peiro is an architect and digital fabrication expert with a Master of Advanced Studies ETH in Architecture and Digital Fabrication from ETH Zurich. His work integrates computational design, robotic fabrication, and sustainable construction, with a strong focus on automation and bio-based materials. He has collaborated as a researcher and developer with the Canadian Pavilion for the Venice Biennale 2025 and has developed several Grasshopper plugins focused on computational design, mathematics, and image processing. Hamid's work has been recognized in both national and international competitions, including awards from Philips Lighting and other design innovation platforms.

Website: Hamidpeiro.com

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Faculty Assistant



Aleksandra Kraeva (Sasha) is an architect and digital technologist in the AEC field, holding a Master's in Robotics and Advanced Construction from IAAC. Her interdisciplinary practice spans Architecture, Robotics R&D, and the design of Interactive Environments. She works at the intersection of design, engineering, and technology, developing methods that integrate robotic fabrication, sensor data, and algorithmic control into spatial systems that are both functional and expressive.

Website / Instagram: @sashakraeva

Linkedin: <https://www.linkedin.com/in/sashanosense/>

Basic schedule: activities and faculty

1. INTRODUCTION + SYSTEM THINKING (Mapping & Assembly Logic) // Hamid Peiro
2. MAPPING STRATEGIES (Input → Rules → Assembly Behavior) // Hamid Peiro
3. FINITE STATE MACHINES (FSM) FOR ASSEMBLY SEQUENCING // Hamid Peiro
4. MODULAR ASSEMBLY DEVELOPMENT (Rules, Connections, Prototyping) //
5. HUMAN-IN-THE-LOOP + PROJECTION FEEDBACK //
6. PROJECT DEVELOPMENT (System Integration & Debugging) //
7. FINAL PRESENTATIONS (System Demo + Assembly Logic Explanation) //

Exercise

Students must choose one specific modular assembly project at the beginning of the course and develop it progressively throughout all sessions.

In the first session, each student (or group) is required to:

- Define a clear assembly concept
- Identify the type of modules and connection logic
- Submit a short project proposal describing their system, rules, and intended workflow

During the course, the project will be incrementally developed by integrating the topics covered in each session:

- Mapping → define relationships between inputs and assembly rules
- Finite State Machine (FSM) → structure assembly sequence and validation logic
- Assembly Logic → refine connections, constraints, and modular behavior
- Human-in-the-loop → introduce user input to control or modulate the system
- Projection / Visualization → provide real-time feedback (valid/invalid states, next steps)

By the end of the course, students must present:

- A fully developed assembly system
- A working interactive prototype
- A clear explanation of their rules, states, and assembly workflow

Deliverables

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Students are requested to submit all the material through the provided Google Form, and publish a Blog Post on the IAAC Blog (blog.iaac.net) for each project (see ["IAAC | Publication Guidelines"](#)) within the maximum of two calendar days after the Final Presentation.

Grading System

- 0 - 4.9 Fail (submission of a supplementary work by January)

- 5.0 - 6.9 Pass
- 7.0 - 8.9 Good
- 9.0 - 10 Excellent/Distinction.

Evaluation, Assessment, and Final Grade

Students will be evaluated according to the following criteria:

- Attendance and participation - 15 %*
- Blog posts and Submissions 20 %*
- Academic performance - 65%

Articulated into:

- Demonstration of imaginative and critical thinking skills.
- Clarity, precision of argument, creativity and depth.
- Development of a project that is appropriate to the criteria developed in class and that of the discipline of architecture, design and engineering.
- Demonstration of technical knowledge essential to the making of architecture.
- Ability to communicate intentions verbally and graphically.
- Understanding of cultural, architectural, technical, and historical ramifications of projects.
- Conscientious effort over the course of the entire quarter.
- Completion of all requirements in a timely manner.
- Participation in individual and group reviews.

**Attendance and completion of submissions are mandatory, students failing to submit or not being present to at least 75% of the classes, without justification included in IAAC Terms & Conditions, will fail.*

Students who submit after the deadlines defined by the faculty and coordination will be subject to penalty and the grade will be automatically lowered by 0.3 point for every day of delay.

Incomplete submission is considered as missing submission.

References / Bibliography

- <https://sutd-cgl.github.io/supp/Publication/projects/2022-EG-AssemblyTutorial/index.html>
- https://www.research-collection.ethz.ch/entities/publication/727d44fb-f4fc-45e0-80b8-865d17f936ac?utm_source=ce
- <https://robotics.utoronto.ca/news/robot-made-2024-digital-fabrication-brings-engineering-and-architecture-students-together>
- <https://www.icd.uni-stuttgart.de/research/#id-752031ef>
- <https://cap-uncertainty.epfl.ch/program>
- Augmented reality for fabrication
- <https://arkite.com/>
- https://www.youtube.com/watch?v=Vcd_vk3n-HY
- References in digital fabrication
- <https://epfl-enac.github.io/MANSLAB-IBOIS-EESD-timberstone/>
- <https://robohub.org/introduction-to-behavior-trees/>
- Rule-based design
- <https://medium.com/@isohale/shape-grammars-1989ddcdeef7>
- <https://www.icd.uni-stuttgart.de/research/research-tools/abxm-framework/>